

A Zero-Dimensional Metric Space with No Bi-Lipschitz Besicovitch Remetrization

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Status: claimed counterexample / full negative answer, likely valid, pending human review.

1 Source question

Le Donne and Rigot ask in Section 8.3 of [2], on printed page 55:

To our knowledge, there is no known example of a metric space (X, d) of finite topological dimension for which there is no distances biLipschitz equivalent to d satisfying BCP.

Equivalently: does finite Lebesgue covering dimension guarantee that the bi-Lipschitz class of a metric contains a distance satisfying the Besicovitch Covering Property?

metric spaces for which BCP does not hold. Hence it is not clear whether one can easily find connections between the validity of BCP and the property of being of negative type.

8.3. Finite topological dimension and an open problem. It is simple to show that if a metric space X satisfies BCP then X has finite topological dimension. Here, topological dimension means Lebesgue covering dimension. Indeed, let N be the constant in the definition of BCP, see Definition 3.1. Without loss of generality we may assume that X is bounded. Then, given an open cover $\mathcal{U}$ of X , we shall prove that there is a refinement of $\mathcal{U}$ with multiplicity at most N , deducing that the topological dimension is at most $N - 1$. For every point a in X take a ball of small enough radius so that it is included in one element of $\mathcal{U}$. By the BCP this family of balls has a subfamily with multiplicity N covering the space X .

If a metric space is assumed to be separable, then by the last argument, together with Remark 3.5, we have that WBCP implies finite topological dimension. With a longer argument, in [Nag64] J. Nagata showed that also WBCP implies finite topological dimension, even if the space is not separable.

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Figure 1: The open problem in Section 8.3 of [2], printed page 55.

2 Definitions

A metric space (Y, δ) satisfies *BCP with constant Q* if every centered family of balls covering a bounded set of centers admits a finite or countable covering subfamily of multiplicity at most Q .

A finite family $\{B_\delta(y_i, r_i)\}_{i=1}^s$ of closed balls is a *Besicovitch family* if its balls have a common point and no ball contains the center of another. The *weak Besicovitch Covering Property* (WBCP) with constant Q says that every such family has cardinality at most Q . BCP with constant Q implies WBCP with the same constant [2, Section 3.1]. We use closed balls in BCP and WBCP, matching the convention of the source; open balls will be denoted by $B_\delta^\circ(y, r)$.

Two metrics d and ρ on the same set are bi-Lipschitz equivalent if there is $L \geq 1$ such that

$$L^{-1}d(x, y) \leq \rho(x, y) \leq Ld(x, y) \quad (x, y \in X).$$

3 Proof intuition

Put increasingly fine finite grids of cubes of *every* dimension into mutually separated components. The resulting space is countable and discrete, hence has topological dimension zero. Nevertheless, any single metric bi-Lipschitz equivalent to it must use one distortion constant on all the grids, and if it satisfied BCP it would use one Besicovitch constant on all of them as well.

Fix a dimension larger than that hypothetical Besicovitch constant. The restricted metrics on the finer and finer grids have a uniformly convergent subsequence. Their limit is a metric on the full cube, bi-Lipschitz to the usual cube metric. WBCP passes to this limit because a Besicovitch family is described by finitely many strict center-exclusion inequalities; slightly enlarging the radii preserves those inequalities on a sufficiently fine grid. Finally, WBCP with constant Q on a compact space yields open-cover refinements of multiplicity Q by a direct equal-radius greedy selection. Thus the limiting cube would have covering dimension at most $Q - 1$, in conflict with its actual dimension.

4 The counterexample

For $m, k \in \mathbb{N}$, $m, k \geq 1$, let

$$D_{m,k} := \left\{0, 2^{-k}, 2 \cdot 2^{-k}, \dots, 1\right\}^m \subset [0, 1]^m.$$

Regard the sets $D_{m,k}$ as pairwise disjoint labelled copies and set

$$X := \bigsqcup_{m,k \geq 1} D_{m,k}.$$

Define a metric d on X by

$$d(x, y) := \begin{cases} \|x - y\|_\infty, & x, y \in D_{m,k} \text{ for the same } (m, k), \\ 3, & x, y \text{ lie in distinct components.} \end{cases}$$

The triangle inequality is immediate because every component has diameter at most 1. Each point is isolated: it lies in a finite component, so its distance from all other points has a positive minimum. Consequently X is a countable discrete topological space and

$$\dim_{\text{top}} X = 0.$$

It is also complete and bounded, though neither extra fact is needed.

Theorem 1. *No metric bi-Lipschitz equivalent to d satisfies the Besicovitch Covering Property. Hence (X, d) is a zero-dimensional counterexample to the question of Le Donne and Rigot.*

We isolate the compactness step.

Lemma 2 (Grid-limit lemma). *Let $K = [0, 1]^m$ with its ℓ_∞ metric, and let $D_k = \{0, 2^{-k}, \dots, 1\}^m$. Suppose δ_k is a metric on D_k such that, for fixed L, Q independent of k ,*

$$L^{-1}\|u - v\|_\infty \leq \delta_k(u, v) \leq L\|u - v\|_\infty,$$

and (D_k, δ_k) satisfies WBCP with constant Q . Then a subsequence converges on approximating grid points to a metric δ_∞ on K which is L -bi-Lipschitz equivalent to $\|\cdot\|_\infty$ and satisfies WBCP with constant Q .

Proof. On $D_k \times D_k$, the function $(u, v) \mapsto \delta_k(u, v)$ satisfies

$$|\delta_k(u, v) - \delta_k(u', v')| \leq L(\|u - u'\|_\infty + \|v - v'\|_\infty).$$

Extend it to $K \times K$ without increasing this Lipschitz constant, for example by the McShane extension formula [3], and clip the result to $[0, L]$. The extensions are uniformly bounded and equicontinuous, so a subsequence converges uniformly to a continuous function $\delta_\infty : K \times K \rightarrow [0, L]$.

If $u_k, v_k \in D_k$ converge to $u, v \in K$ along this subsequence, uniform convergence and equicontinuity give

$$\delta_k(u_k, v_k) \longrightarrow \delta_\infty(u, v).$$

Approximating two or three arbitrary points of K by grid points now passes symmetry, the triangle inequality, and the two bi-Lipschitz bounds to the limit. The lower bound separates distinct points, so δ_∞ is a metric.

It remains to pass WBCP. Let $\{B_{\delta_\infty}(x_i, r_i)\}_{i=1}^s$ be a Besicovitch family and choose x_0 in the intersection of all its balls. Thus

$$\delta_\infty(x_0, x_i) \leq r_i, \quad \delta_\infty(x_i, x_j) > \max\{r_i, r_j\} \quad (i \neq j).$$

Choose $\varepsilon > 0$ so small that

$$\delta_\infty(x_i, x_j) > \max\{r_i, r_j\} + 3\varepsilon \quad (i \neq j).$$

Choose grid approximants $x_{i,k} \rightarrow x_i$, including $i = 0$. For all large k , convergence of the finitely many relevant distances shows that the balls

$$B_{\delta_k}(x_{i,k}, r_i + \varepsilon), \quad i = 1, \dots, s,$$

have common point $x_{0,k}$ and contain none of each other's centers. They form a Besicovitch family in (D_k, δ_k) , hence $s \leq Q$. Therefore δ_∞ satisfies WBCP with constant Q . $\square$

Lemma 3 (Compact WBCP dimension bound). *If a compact metric space (K, δ) satisfies WBCP with constant Q , then its Lebesgue covering dimension is at most $Q - 1$.*

Proof. Let $\mathcal{U}$ be an open cover of K and let $\lambda > 0$ be a Lebesgue number. Choose $r > 0$ with $4r < \lambda$. Greedily choose centers $c_1, \dots, c_t$ so that each new center is outside all previously selected closed r -balls. Compactness makes the process finite, since the centers are r -separated. A maximal such family of closed r -balls covers K , and $\delta(c_i, c_j) > r$ for $i \neq j$.

Since the family is finite, choose $\eta > 0$ so small that

$$r + \eta < \min_{i \neq j} \delta(c_i, c_j) \quad \text{and} \quad 2(r + \eta) < \lambda,$$

with the first condition omitted if $t = 1$. The open balls $B_\delta^\circ(c_i, r + \eta)$ still cover K and refine $\mathcal{U}$. To bound their multiplicity, fix z and let I index the open balls that contain z . For every $i \in I$, choose

$$\delta(z, c_i) < s_i < r + \eta.$$

The closed balls $B_\delta(c_i, s_i)$, $i \in I$, have common point z , while $\delta(c_i, c_j) > r + \eta > \max\{s_i, s_j\}$ for distinct indices. They form a Besicovitch family, so $|I| \leq Q$. Thus the open refinement has multiplicity at most Q , and $\dim_{\text{top}} K \leq Q - 1$. $\square$

Proof of Theorem 1. Suppose, toward a contradiction, that a metric ρ on X is L -bi-Lipschitz equivalent to d and satisfies BCP with an integer constant Q . BCP implies WBCP with constant Q . A Besicovitch family whose centers and common point lie in one component $D_{m,k}$ is also a Besicovitch family in the ambient space, because the defining distance inequalities are unchanged. Therefore every restriction $\rho|_{D_{m,k}}$ satisfies WBCP with constant Q .

Fix $m = Q + 1$ and write $\delta_k = \rho|_{D_{m,k}}$ after identifying the labelled component with its grid in $[0, 1]^m$. The bi-Lipschitz inequalities give

$$L^{-1}\|u - v\|_\infty \leq \delta_k(u, v) \leq L\|u - v\|_\infty.$$

Lemma 2 produces on $[0, 1]^m$ an L -bi-Lipschitz equivalent metric δ_∞ satisfying WBCP with constant Q . The bi-Lipschitz bounds give the usual topology, while Lemma 3 gives

$$m = \dim_{\text{top}}[0, 1]^m \leq Q - 1.$$

This contradicts $m = Q + 1$. No such ρ exists. $\square$

5 Verification notes

- The construction is topologically discrete even though the within-block mesh tends to zero: every individual point belongs to one finite block.
- The proof uses only the two constants attached to one hypothetical remetrization, namely its global bi-Lipschitz constant L and global BCP constant Q .
- Restricting WBCP to a component is legitimate even though an ambient ball may contain points in other components; the common point and all center-exclusion inequalities already occur inside the component.
- The grid-limit argument does not assume that BCP itself is closed under limits. It passes only WBCP, whose obstruction is a finite collection of strict inequalities. This is the same stability mechanism used in [2, Lemma 7.7], here adapted to varying finite grids.
- The final dimension estimate is proved directly for compact spaces and does not require doubling, completeness of the hypothetical metric, or a measure-differentiation theorem.

6 Limitations and novelty check

This answers the source question negatively in the class of complete bounded countable metric spaces, indeed with a topologically discrete example. It does not produce a connected, locally connected, geodesic, or doubling counterexample; those stronger variants remain untouched.

On July 21, 2026, the run registry, solution, attempt, and proof-gap indexes were searched for arXiv:1512.04936 and the core BCP/bi-Lipschitz/topological dimension terminology. Bounded web and arXiv searches used the exact source sentence and close variants involving “finite topological dimension”, “bi-Lipschitz equivalent”, and “Besicovitch Covering Property”. They found the source paper, its antecedent Heisenberg-group paper, Aldaz’s 2022 survey of Besicovitch and doubling properties, and adjacent work, but no later paper claiming this counterexample or resolving the exact remetrization question. This is a bounded novelty check, not a guarantee of originality.

7 Human-review recommendation

The highest-value review is the grid-limit lemma: verify the uniform extension argument and the $+\varepsilon$ transfer of an arbitrary Besicovitch family to a sufficiently fine grid. Next check that ambient WBCP restricts to each labelled component exactly as used. The remaining steps are the standard covering-dimension characterization and the classical identity $\dim[0, 1]^m = m$ [1].

References

- [1] W. Hurewicz and H. Wallman, *Dimension Theory*, Princeton University Press, 1941.
- [2] E. Le Donne and S. Rigot, *Besicovitch Covering Property on graded groups and applications to measure differentiation*, arXiv:1512.04936v1 (2015).
- [3] E. J. McShane, *Extension of range of functions*, Bulletin of the American Mathematical Society **40** (1934), 837–842.